

ECE238L Computer Logic Design

Lab 4: D-Latch and D-Flip-Flop

Section 003: Wednesday 2:00-4:45

Name: Tomas Salaz

D-Latch VHDL Source Code, Simulation Code and Constraints

D_Latch.vhd

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-- Company:
-- Engineer:
-- Create Date: 03/04/2025 01:08:37 PM
-- Design Name:
-- Module Name: D_Latch - Behavioral
-- Project Name:
-- Target Devices:
-- Tool Versions:
-- Description:
-- Dependencies:
-- Revision:
-- Revision 0.01 - File Created
-- Additional Comments:
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--
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;

-- Uncomment the following library declaration if using
-- arithmetic functions with Signed or Unsigned values
--use IEEE.NUMERIC_STD.ALL;

-- Uncomment the following library declaration if instantiat
-- any Xilinx leaf cells in this code.
--library UNISIM;
--use UNISIM.VComponents.all;

entity D_Latch is
    Port (
        D : in STD_LOGIC;
        CLK : in STD_LOGIC;
        Q : out STD_LOGIC);
end D_Latch;

architecture Behavioral of D_Latch is
begin
    state : process (D, CLK)
    begin
        if CLK = '1' then
            Q <= D;
        --else
        -- Do Nothing
        end if;
    end process;
end Behavioral;
```

D_Latch_tb.vhd

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-- Description:
-- Dependencies:
-- Revision:
-- Revision 0.01 - File Created
-- Additional Comments:
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--
--
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;

-- Uncomment the following library declaration if using
-- arithmetic functions with Signed or Unsigned values
--use IEEE.NUMERIC_STD.ALL;

-- Uncomment the following library declaration if instantiat
-- any Xilinx leaf cells in this code.
--library UNISIM;
--use UNISIM.VComponents.all;

entity D_Latch_tb is
    Port (
        D : in STD_LOGIC;
        CLK : in STD_LOGIC;
        Q : out STD_LOGIC);
end D_Latch_tb;

architecture bench of D_Latch_tb is
    component D_Latch is
        Port (
            D : in STD_LOGIC;
            CLK : in STD_LOGIC;
            Q : out STD_LOGIC);
    end component;

    signal D_tb, CLK_tb, Q_tb : STD_LOGIC;
    constant CLK_period : time := 10 ns;

begin
    uut : D_Latch port map
        ( D => D_tb,
          CLK => CLK_tb,
          Q => Q_tb);

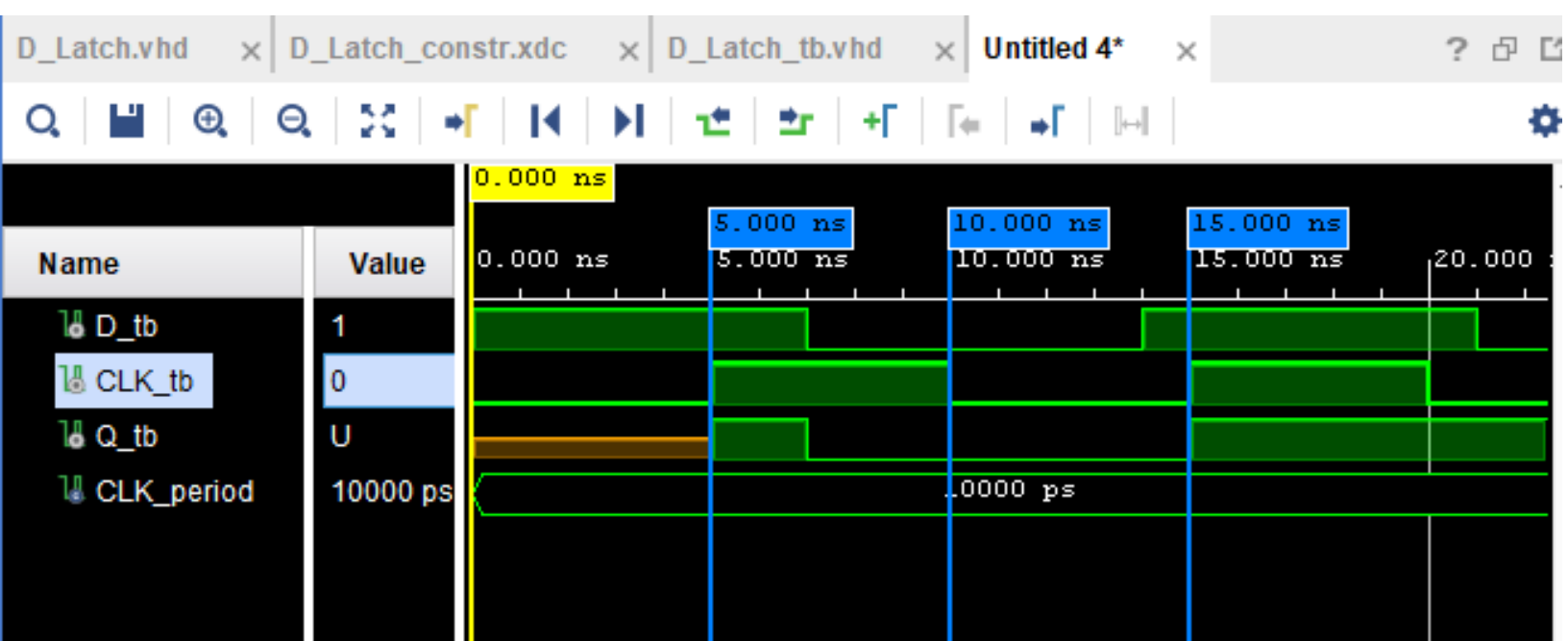
    CLK_process : process
    begin
        CLK_tb <= '0'; wait for CLK_period/2;
        CLK_tb <= '1'; wait for CLK_period/2;
    end process;

    stimulus : process
    begin
        D_tb <= '1'; wait for 7 ns;
        D_tb <= '0'; wait for 7 ns;
    end process;
end bench;
```

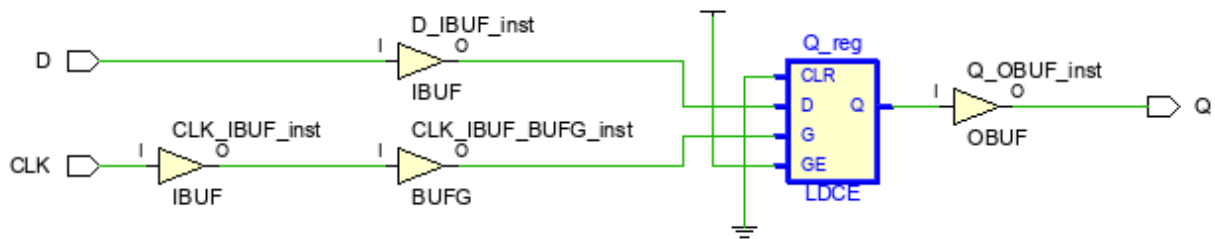
D_Latch_constr.xdc

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set_property IOSTANDARD LVCMOS33 [get_ports D]
set_property IOSTANDARD LVCMOS33 [get_ports Q]
set_property IOSTANDARD LVCMOS33 [get_ports CLK]
set_property PACKAGE_PIN L16 [get_ports D]
set_property PACKAGE_PIN H17 [get_ports Q]
set_property PACKAGE_PIN J15 [get_ports CLK]
set_property CLOCK_DEDICATED_ROUTE FALSE [get_nets CLK_IBUF]
```

D-Latch Waveform



D-Latch Schematic



D-Flip-Flop VHDL Source Code, Simulation Code and Constraints

D_Flip_Flop.vhd

```
1  -- Company:
2  -- Engineer:
3  --
4  --
5  -- Create Date: 03/04/2025 01:46:13 PM
6  -- Design Name:
7  -- Module Name: D_Flip_Flop - Behavioral
8  -- Project Name:
9  -- Target Devices:
10 -- Tool Versions:
11 -- Description:
12 --
13 -- Dependencies:
14 --
15 -- Revision:
16 -- Revision 0.01 - File Created
17 -- Additional Comments:
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21
22 library IEEE;
23 use IEEE.STD_LOGIC_1164.ALL;
24
25 -- Uncomment the following library declaration if using
26 -- arithmetic functions with Signed or Unsigned values
27 --use IEEE.NUMERIC_STD.ALL;
28
29 -- Uncomment the following library declaration if instantiating
30 -- any Xilinx leaf cells in this code.
31 --library UNISIM;
32 --use UNISIM.VComponents.all;
33
34 entity D_Flip_Flop is
35     Port (
36         D : in STD_LOGIC;
37         CLK : in STD_LOGIC;
38         Q : out STD_LOGIC);
39 end D_Flip_Flop;
40
41 architecture Behavioral of D_Flip_Flop is
42     begin
43         state : process (CLK)
44             begin
45                 if CLK'event and CLK = '1'
46                     then Q <= D;
47                 --else
48                 -- do nothing
49                 end if;
50             end process;
51 end Behavioral;
```

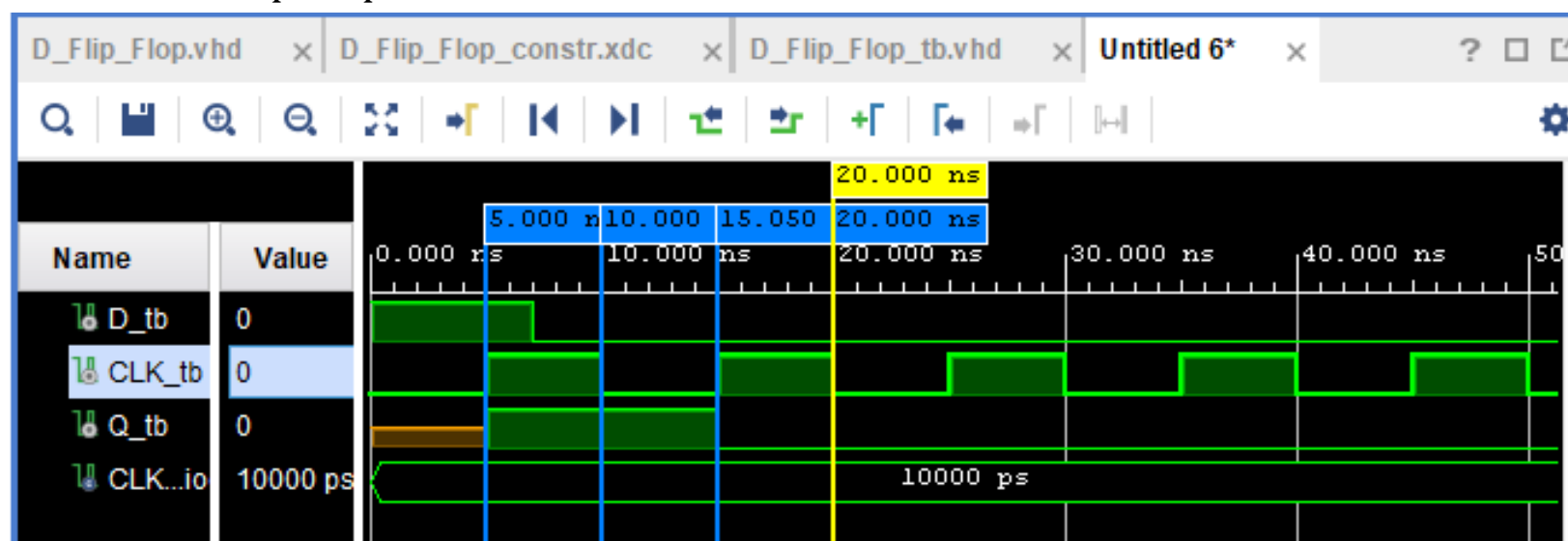
D_Flip_Flop_tb.vhd

```
12 --
13 -- Dependencies:
14 --
15 -- Revision:
16 -- Revision 0.01 - File Created
17 -- Additional Comments:
18 --
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21
22 library IEEE;
23 use IEEE.STD_LOGIC_1164.ALL;
24
25 -- Uncomment the following library declaration if using
26 -- arithmetic functions with Signed or Unsigned values
27 --use IEEE.NUMERIC_STD.ALL;
28
29 -- Uncomment the following library declaration if instantiating
30 -- any Xilinx leaf cells in this code.
31 --library UNISIM;
32 --use UNISIM.VComponents.all;
33
34 entity D_Flip_Flop_tb is
35     -- Port ( );
36 end D_Flip_Flop_tb;
37
38 architecture bench of D_Flip_Flop_tb is
39     component D_Flip_Flop is
40         Port (
41             D : in STD_LOGIC;
42             CLK : in STD_LOGIC;
43             Q : out STD_LOGIC);
44     end component;
45
46     signal D_tb, CLK_tb, Q_tb : STD_LOGIC;
47     constant CLK_period : time := 10 ns;
48
49     begin
50         uut: D_Flip_Flop port map (
51             D => D_tb,
52             CLK => CLK_tb,
53             Q => Q_tb);
54
55         CLK_process: process
56             begin
57                 CLK_tb <= '0'; wait for CLK_period/2;
58                 CLK_tb <= '1'; wait for CLK_period/2;
59             end process;
60
61         stimulus: process
62             begin
63                 D_tb <= '1'; wait for 7 ns;
64                 D_tb <= '0'; wait for 7 ns;
65                 wait;
66             end process;
67     end bench;
```

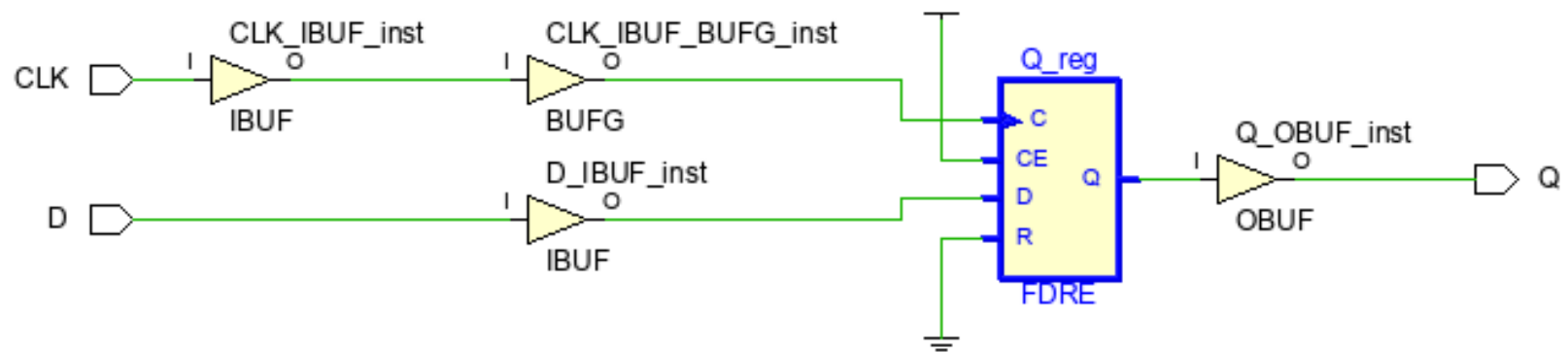
D_Flip_Flop_constr.xdc

```
1 set_property IOSTANDARD LVCMOS33 [get_ports CLK]
2 set_property IOSTANDARD LVCMOS33 [get_ports D]
3 set_property IOSTANDARD LVCMOS33 [get_ports Q]
4 set_property PACKAGE_PIN J15 [get_ports CLK]
5 set_property PACKAGE_PIN L16 [get_ports D]
6 set_property PACKAGE_PIN H17 [get_ports Q]
7 set_property CLOCK_DEDICATED_ROUTE FALSE [get_nets CLK_IBUF]
```

D-Flip-Flop Waveform



D-Flip-Flop Schematic



Register VHDL Source Code, Simulation Code and Constraints

Lab4C.vhd

C:/Users/salaz25/ECE238/Lab4C/Lab4C.srcs/sources_1/new/Lab4C.vhd

```
1  -- Company:
2  -- Engineer:
3
4  -- Create Date: 03/04/2025 02:10:37 PM
5  -- Design Name:
6  -- Module Name: Lab4C - Behavioral
7  -- Project Name:
8  -- Target Devices:
9  -- Tool Versions:
10 -- Description:
11 -- Dependencies:
12
13 -- Revision:
14 -- Revision 0.01 - File Created
15 -- Additional Comments:
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20
21
22 library IEEE;
23 use IEEE.STD_LOGIC_1164.ALL;
24
25 -- Uncomment the following library declaration if using
26 -- arithmetic functions with Signed or Unsigned values
27 --use IEEE.NUMERIC_STD.ALL;
28
29 -- Uncomment the following library declaration if instantiating
30 -- any Xilinx leaf cells in this code.
31 --library UNISIM;
32 --use UNISIM.VComponents.all;
33
34
35 entity Lab4C is
36   Port ( CLK : in STD_LOGIC;
37         D : in STD_LOGIC_VECTOR(7 downto 0);
38         Q : out STD_LOGIC_VECTOR(7 downto 0));
39 end Lab4C;
40
41 architecture Behavioral of Lab4C is
42 begin
43   process (CLK)
44   begin
45     if rising_edge(CLK) then
46       Q <= D;
47     end if;
48   end process;
49 end Behavioral;
```

Lab4C_tb.vhd

C:/Users/salaz25/ECE238/Lab4C/Lab4C.srcs/sim_1/new/Lab4C_tb.vhd

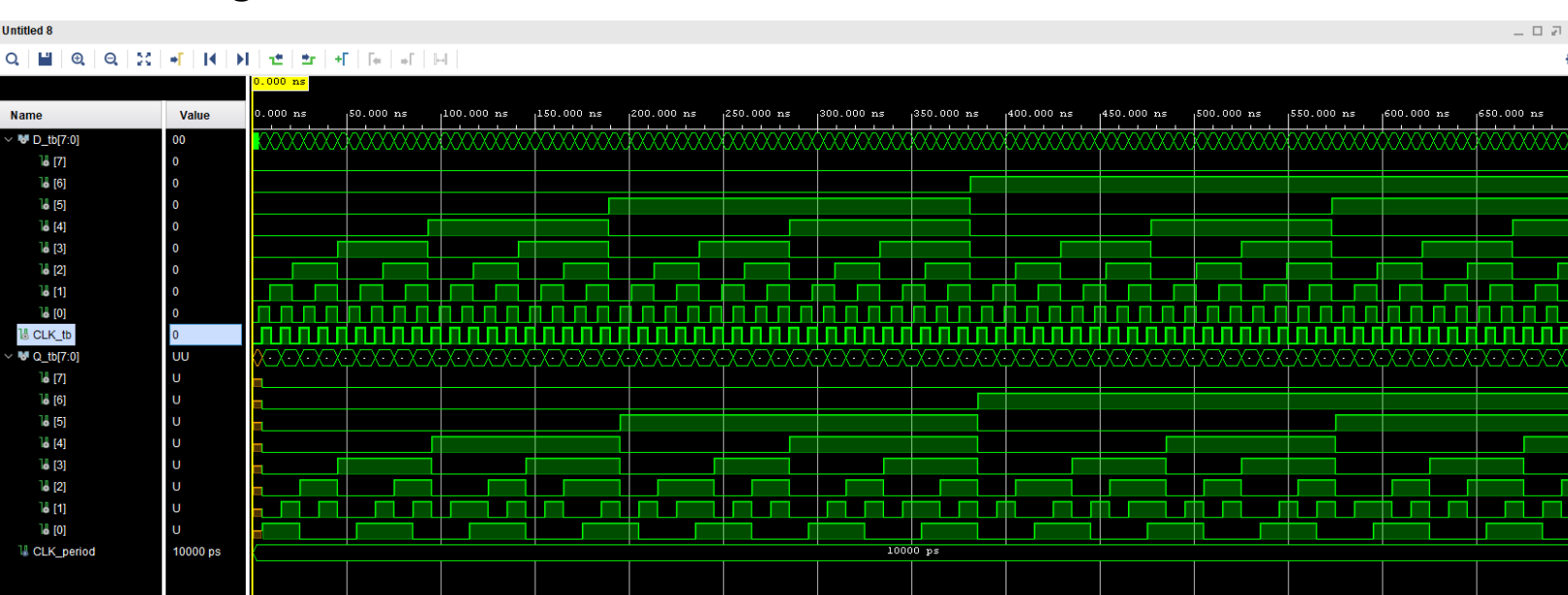
```
17 -- Additional Comments:
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21
22 library IEEE;
23 use IEEE.STD_LOGIC_1164.ALL;
24 use IEEE.numeric_std.ALL;
25
26 -- Uncomment the following library declaration if using
27 -- arithmetic functions with Signed or Unsigned values
28 --use IEEE.NUMERIC_STD.ALL;
29
30 -- Uncomment the following library declaration if instantiating
31 -- any Xilinx leaf cells in this code.
32 --library UNISIM;
33 --use UNISIM.VComponents.all;
34
35 entity Lab4C_tb is
36   Port ( );
37 end Lab4C_tb;
38
39 architecture bench of Lab4C_tb is
40   component Lab4C is
41     Port ( CLK : in STD_LOGIC;
42           D : in STD_LOGIC_VECTOR(7 downto 0);
43           Q : out STD_LOGIC_VECTOR(7 downto 0));
44   end component;
45
46   signal D_tb : STD_LOGIC_VECTOR(7 downto 0);
47   signal CLK_tb : STD_LOGIC;
48   signal Q_tb : STD_LOGIC_VECTOR(7 downto 0);
49   constant CLK_period : time := 10 ns;
50 begin
51   uut : Lab4C port map (
52     CLK => CLK_tb,
53     D => D_tb,
54     Q => Q_tb);
55
56   CLK_process: process
57   begin
58     CLK_tb <= '0'; wait for CLK_period/2;
59     CLK_tb <= '1'; wait for CLK_period/2;
60   end process;
61
62   stimulus: process
63   begin
64     D_tb <= "00000000";
65     for i in 0 to 511 loop
66       wait for 3 ns; D_tb <= STD_LOGIC_VECTOR(unsigned(D_tb)+1);
67     end loop;
68   end process;
69 end bench;
```

Lab4C_constr.xdc

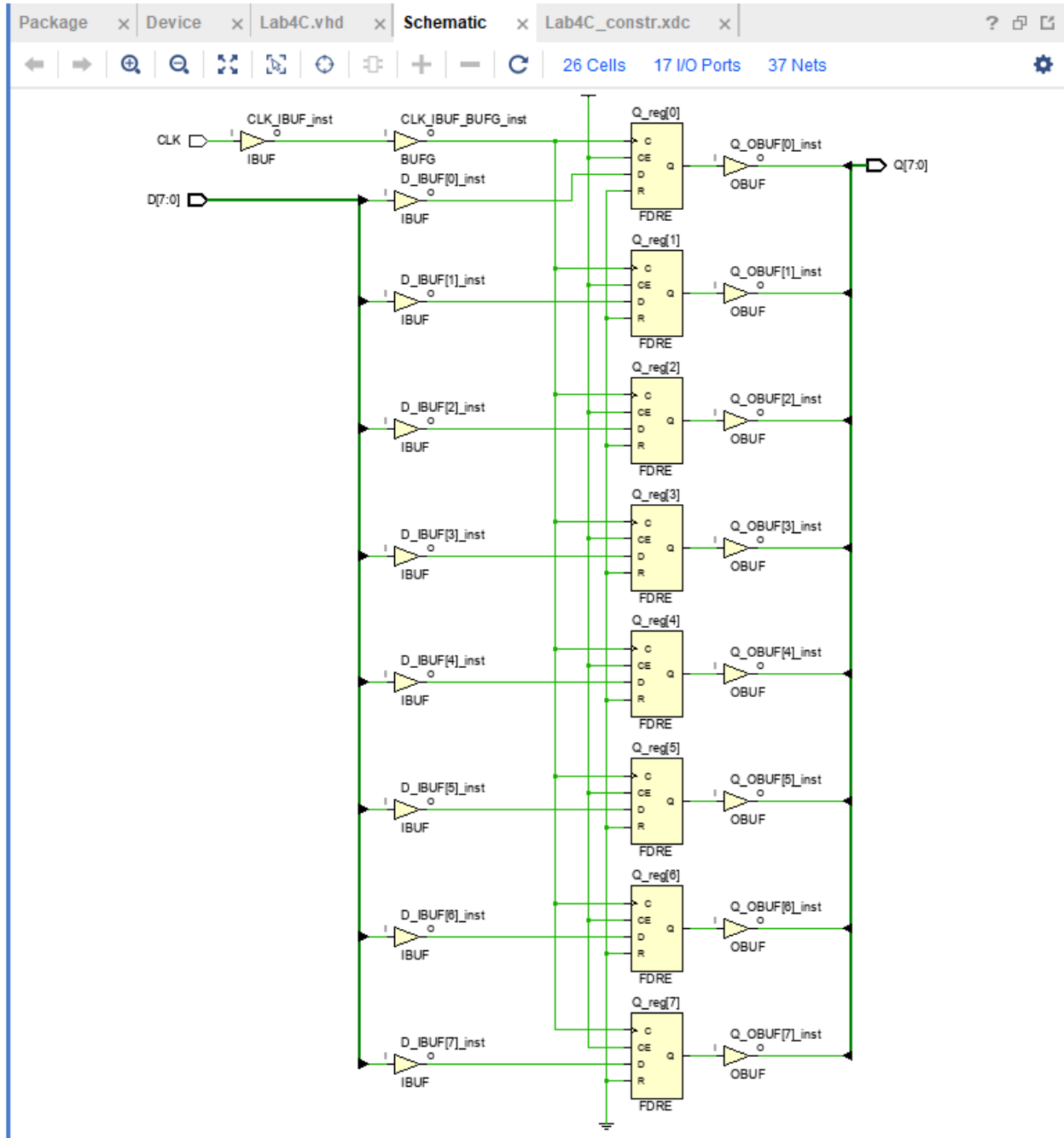
C:/Users/salaz25/ECE238/Lab4C/Lab4C.srcs/constrs_1/new/Lab4C_constr.xdc

```
1 set_property IOSTANDARD LVCMOS33 [get_ports {D[7]}]
2 set_property IOSTANDARD LVCMOS33 [get_ports {D[6]}]
3 set_property IOSTANDARD LVCMOS33 [get_ports {D[5]}]
4 set_property IOSTANDARD LVCMOS33 [get_ports {D[4]}]
5 set_property IOSTANDARD LVCMOS33 [get_ports {D[3]}]
6 set_property IOSTANDARD LVCMOS33 [get_ports {D[2]}]
7 set_property IOSTANDARD LVCMOS33 [get_ports {D[1]}]
8 set_property IOSTANDARD LVCMOS33 [get_ports {D[0]}]
9 set_property IOSTANDARD LVCMOS33 [get_ports {Q[7]}]
10 set_property IOSTANDARD LVCMOS33 [get_ports {Q[6]}]
11 set_property IOSTANDARD LVCMOS33 [get_ports {Q[5]}]
12 set_property IOSTANDARD LVCMOS33 [get_ports {Q[4]}]
13 set_property IOSTANDARD LVCMOS33 [get_ports {Q[3]}]
14 set_property IOSTANDARD LVCMOS33 [get_ports {Q[2]}]
15 set_property IOSTANDARD LVCMOS33 [get_ports {Q[1]}]
16 set_property IOSTANDARD LVCMOS33 [get_ports {Q[0]}]
17 set_property IOSTANDARD LVCMOS33 [get_ports {CLK}]
18 set_property PACKAGE_PIN J15 [get_ports CLK]
19 set_property PACKAGE_PIN L16 [get_ports {D[7]}]
20 set_property PACKAGE_PIN M13 [get_ports {D[6]}]
21 set_property PACKAGE_PIN R15 [get_ports {D[5]}]
22 set_property PACKAGE_PIN R17 [get_ports {D[4]}]
23 set_property PACKAGE_PIN T18 [get_ports {D[3]}]
24 set_property PACKAGE_PIN U18 [get_ports {D[2]}]
25 set_property PACKAGE_PIN R13 [get_ports {D[1]}]
26 set_property PACKAGE_PIN T8 [get_ports {D[0]}]
27 set_property PACKAGE_PIN K15 [get_ports {Q[7]}]
28 set_property PACKAGE_PIN J13 [get_ports {Q[6]}]
29 set_property PACKAGE_PIN M14 [get_ports {Q[5]}]
30 set_property PACKAGE_PIN R18 [get_ports {Q[4]}]
31 set_property PACKAGE_PIN V17 [get_ports {Q[3]}]
32 set_property PACKAGE_PIN U17 [get_ports {Q[2]}]
33 set_property PACKAGE_PIN U16 [get_ports {Q[1]}]
34 set_property PACKAGE_PIN V16 [get_ports {Q[0]}]
35 set_property CLOCK_DEDICATED_ROUTE FALSE [get_nets CLK_IBUF]
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Register Waveform



Register Schematic



Summary:

For this Lab we had to make 2 low-level entities and implement them in a high-level entity. The 2 lower-level entities were D-Latch and D-Flip-Flop. The D-Latch is used to change the output (Q) depending on if the input signal (D) is high or low. D-Flip-Flop is similar but instead of 'latching' the output (Q), it Flips them, and is still based on the input (D).

The last part was the Register. This File implemented both the D-Latch and D-Flip-Flop. The functionality was that we have 8 bit output (Q), and the D input (high or low). We can change the output when the rising edge happens to reduce the 'noise'.

Problems:

This Lab seemed like a step down in difficulty and was fairly easy to understand. My biggerst problem was that I didn't understand the Rising Edge and the Falling Edge until the TA (Matthew) explained it to me.

Hints:

Like I said before this lab was not too hard so I really don't have any hints.

Recommendations:

None, Good Lab